

鄭聖文 Sheng-Wen (Colin) Cheng

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Employment

NVIDIA System Software Engineer	Feb. 2024 - Present Taipei, Taiwan
Avilon Intelligence Embedded Systems Engineer	Sept. 2018 - Mar. 2021 Tainan, Taiwan

Education

The University of Texas at Austin M.S. in Computer Science	Aug. 2025 - Dec. 2026 Texas, USA
National Taiwan University Ph.D. Student, Electrical Engineering	Sept. 2022 - May 2024 Taipei, Taiwan
National Yang Ming Chiao Tung University M.S. in Robotics	Sept. 2019 - Nov. 2021 Hsinchu, Taiwan
Providence University B.E. in Computer Science and Information Engineering	Sept. 2015 - June 2019 Taichung, Taiwan

Academic Service

- Invited Reviewer for the **American Control Conference (ACC 2026)**

Certificate

MITx MicroMasters Program in Statistics and Data Science Certificate Program by Massachusetts Institute of Technology [Program Letter]	Jan. 2024 – Dec. 2025 MITx
• Coursework: Probability, Statistics, Machine Learning, Time Series Analysis, and proctored Capstone Exam.	

Projects

NCRL Flight Control [Video]
• An agile quadrotor system with nonlinear control and navigation, licensed to the Taiwan Space Agency (TASA).
Tenok: A Linux-like RTOS for Robotics and IoT [GitHub]
• Developed POSIX-compliant RTOS optimized for ARM Cortex-M with Linux-inspired kernel and Unix features.
PU-01: Twin-boom Fixed-wing UAV [Link]
• Designed and fabricated airframe: 3D modeling, airfoil selection, and aerodynamics/stability analysis with XFLR5.
Semu: Minimalist RISC-V System Emulator [GitHub]
• Contributed to hardware virtualization of GPU, input devices, storage, and random number generator with VirtIO.

Publications

- [1] **S.-W. Cheng** and T.-H. Cheng, “Data-Driven Estimation of Quadrotor Motor Efficiency via Residual Minimization,” *American Control Conference (ACC)*, New Orleans, USA, 2026. [arXiv:2510.11388](#)

- [2] **S.-W. Cheng** and Y.-H. Huang, “A Computationally Efficient GNSS/INS Design of Multirotor based on Error-state Kalman Filter,” *62nd Annual Conference of the Society of Instrument and Control Engineers of Japan (SICE)*, Tsu, Japan, 2023. [\[Link\]](#)
- [3] **S.-W. Cheng** and H.-A. Hung, “Robust State-Feedback H_∞ Control of Quadrotor,” *International Automatic Control Conference (CACs)*, Kaohsiung, Taiwan, 2022. [\[Link\]](#)
- [4] S.-W. Wang, **S.-W. Cheng**, and C.-C. Huang, “Puyuma: Linux-based RTOS Experimental Platform for Constructing Self-Driving Miniature Vehicles,” *Science and Information Conference (SAI)*, London, United Kingdom, 2018. [\[Link\]](#)

Presentations

- [1] “Demystifying VirtIO-GPU: Building a Graphics Virtualization Bridge from Scratch,” Open Source Summit North America (OSSNA 2026, **Linux Foundation Event**), Minneapolis, USA, 2026. [\[Link\]](#) [\[PDF\]](#)
- [2] “Building a Quadrotor Simulator with Python - Modeling, Simulation, and Control,” Open Tech Conference (OpenTechConf 2025), Hong Kong, 2025. [\[Link\]](#) [\[PDF\]](#)
- [3] “Crafting a Vision-Aided Software Stack for UAV,” Embedded Open Source Summit (EOSS 2024, **Linux Foundation Event**), Seattle, USA, 2024. [\[Link\]](#) [\[PDF\]](#)
- [4] “Creating a Linux-like Real-Time Operating System for Quadrotor Drones,” Conference for Open Source Coders, Users, and Promoters (COSCUP 2024), Taiwan, 2024. [\[Link\]](#) [\[PDF\]](#)
- [5] “Tenok: Build a real-time operating system for Robotics,” Conference for Open Source Coders, Users, and Promoters (COSCUP 2023), Taiwan, 2023. [\[Link\]](#) [\[PDF\]](#)

Invited Talks

- [1] “Trends in Machine Learning for Unmanned Aerial Vehicle Applications,” Mobile Open Platform (MOPCON 2024), **Keynote speaker**, Taiwan, 2024. [\[Link\]](#) [\[PDF\]](#)
- [2] PEGATRON Corporation: “Trends and lessons learned in deep learning and generative AI applications for UAV,” Taipei, Taiwan, 2024.